

Buckeye Experimental Aeronautics (BExA) is a student-led engineering organization focused on the design, development, and flight testing of high-speed, jet-powered, remotely piloted aircraft. The organization is built as a multidisciplinary, long-term engineering effort that brings together aerodynamics, propulsion, structures, controls, avionics, and autonomous systems into a unified engineering program.

BExA's mission is to create a structured and technically rigorous environment where students can design and operate increasingly advanced aircraft systems, preparing them for real-world aerospace engineering. The team's near-term objective is a Guinness World Record in high-speed flight, but this goal is intentionally embedded within a broader framework of developing autonomous capabilities, high-speed flight systems, and propulsion testing.

The program follows a staged development roadmap that prioritizes controlled progression and risk management. Initial efforts focus on electric ducted fan (EDF) aircraft for low-risk flight testing and autonomy development, alongside a parallel microturbine ground test campaign to build propulsion expertise. These efforts converge into a turbine-powered demonstrator, ultimately leading to a specialized high-speed platform designed for record-level performance. This progression ensures that technical capability, safety practices, and operational discipline mature together.

Autonomy is introduced early through lower-risk test platforms, allowing the team to mature flight software, sensing, telemetry, fault handling, and repeatable test procedures before carrying those capabilities into higher-speed aircraft. In later platforms, autonomy will support mission execution, flight envelope protection, fault response, geofencing, and data collection while maintaining human oversight.

BExA is structured to operate as a durable engineering program rather than a short-term project. The organization includes both technical and operational divisions that support continuity, documentation, and long-term sustainability of the program. The team is advised by Dr. Matthew H. McCrink, Assistant Professor in Mechanical and Aerospace Engineering, providing technical mentorship and institutional continuity.

Beyond the technical goals, BExA is designed to contribute to Ohio State's broader engineering ecosystem. The program emphasizes interdisciplinary collaboration, hands-on experience with complex systems, and the development of professional engineering practices including design reviews, safety documentation, and structured testing. BExA aims to establish itself as a distinctive MAE program that prepares students to operate at a high level in both industry and research environments.